

Machine Learning Fundamentals

Choose the correct learning problem before choosing an algorithm.



WHY THIS MATTERS

Start from what you already know

Features, targets, tables, distributions, and basic mathematical functions

- 01 Explain supervised and unsupervised learning in plain language.
- 02 Use regression and classification in a small worked example.
- 03 Complete the guided lab and verify generalization.

CORE CONCEPT

The vocabulary of this topic

Choose the correct learning problem before choosing an algorithm.

supervised and unsupervised learning: Supervised learning uses known targets; unsupervised learning searches for structure without target labels.

regression and classification: Regression predicts a continuous quantity; classification predicts a category or category probability.

training and inference: Training estimates model parameters from examples; inference applies fixed learned parameters to new inputs.

generalization: Generalization is useful performance on representative examples that did not influence model fitting.

FOUR CONNECTED IDEAS

How the concept works

01

**supervised and
unsupervised
learning**



02

**regression and
classification**



03

**training and
inference**



04

generalization

A model learns a mapping or structure from training data and is judged on unseen examples that represent future use.

CORE CONCEPT

Read every part of the mechanism

A model learns a mapping or structure from training data and is judged on unseen examples that represent future use.

feature: An input available at prediction time.

target: The value a supervised model learns to predict.

label: A categorical target.

WORKED EXAMPLE

Worked example by hand

Classify house price as regression, spam detection as classification, and customer grouping as clustering.

01

Known values

02

Apply the mechanism

03

Interpret the result

RUN IT AND INSPECT THE RESULT

Map the concept to Python

```
problems = {"price": "regression", "spam": "classification", "segments": "clustering"}  
for name, kind in problems.items():  
    print(name, kind)
```

OUTPUT

```
price regression  
spam classification  
segments clustering
```

Each line corresponds to a concept already explained.

QUALITY GATE

Build the complete mental model

- ☒ Regression targets are continuous; classification targets are categories; clustering has no supplied target.
- ☒ Training learns parameters, validation selects choices, and testing estimates final generalization.
- ☒ Underfitting misses structure; overfitting learns training-specific noise.
- ☒ Leakage occurs whenever training receives information unavailable at real prediction time.

FOUR CONNECTED IDEAS

Notebook readiness map

01

**supervised and
unsupervised
learning**



02

**regression and
classification**



03

**training and
inference**



04

generalization

Use the concepts from this lesson to read and complete
<https://github.com/curiously/AI-Bootcamp/blob/master/03.model-development.ipynb>.

RISK REVIEW

Common beginner mistakes

01 **Calling every prediction problem classification.**

02 **Evaluating on rows that influenced training or preprocessing.**

GUIDED LAB

Frame five business questions by identifying observation, features, target, learning type, and evaluation evidence.

01

02

03

04

Predict

Run

Inspect

Explain

SESSION SYNTHESIS

Keep the system, not isolated definitions.

- 01 Explain supervised and unsupervised learning in plain language.
- 02 Use regression and classification in a small worked example.
- 03 Complete the guided lab and verify generalization.

NEXT EVIDENCE

A problem-framing worksheet with justified learning types and train/validation/test boundaries.